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Review and Analysis of Autonomous Inflow Control Devices for Gas Production Regularization in an Offshore Carbonate Reservoir

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Abstract

Autonomous inflow control devices (AICDs) have been piloted/deployed as one of well lower completions to regulate gas production from the target giant carbonate oil field located offshore Abu Dhabi. The AICDs were introduced to industry for autonomous regularization of gas/water production as well as achievement of uniformly distributed inflow profiles. However, because their performance depends on reservoir/fluid characteristics, further implementation of this technology in the subject field requires review and analysis of actual performance of the piloted/deployed AICDs.

This study presents a series of reviews/analyses on the performances of multiple pilot wells completed with the AICDs. The reviews evaluated well inflow performance and AICDs' nozzle performance (by integrating results of flow tests, pressure build-up tests, and production logging tool (PLT) surveys), which enabled us to calibrate key parameters of the published AICD models via history-matching with numerical well/reservoir models. Subsequently, reservoir simulation studies investigated expected future performance of the AICDs under i) gas breakthrough from horizontal direction and ii) gas coning from overlying gas cap with the calibrated parameters and a multi-segment well capability prepared in commercial reservoir simulators.

The data review of pilot wells showed existence of induced pressure drop through the AICD nozzles as evidenced by time-lapse PLTs in one of the wells though the pressure drop seems lower than designed one. The inflow/pressure matching against PLT data suggested sets of values for the parameters in the AICD model although the parameter related with gas/water contribution has higher uncertainty because of no gas/water breakthrough in the subject well. A hypothetical horizontal gas breakthrough scenario with a refined single well reservoir simulation suggests that AICD installation makes the well life longer by couple of months even with 6-10 times larger drawdown than base case. Subsequently, the full-field simulations (for gas coning scenarios) in planning of future wells demonstrated equalized inflow distribution along the horizontal drain by delaying the gas coning but it highly depends on depth/distance from the gas cap. Selective regulation of the gas production was not anticipated with the parameters used under reasonable range of the gas-oil ratio in the subject field. It should be noted that nozzle leakage scenario was also a

reasonable hypothesis to reproduce the observed PLT data, but it led to massive inflow from these nozzles causing acceleration of gas coning even worse than no AICD case.

It should be noted that limited number of works has been presented focusing on actual long-term AICD performance and its integration for future forecasting. Therefore, this work is intended to serve as a reference to provide diagnostic results of such an advanced lower completion device with remaining challenges and potential risk-mitigation plans.

Introduction

Downhole flow control devices have been used as a part of well completions to control and optimize well performance since 1990s (Al-Khelaiwi and Davies 2007; Li et al. 2011a). Historically, there are two types of downhole flow control devices: inflow control devices (ICDs) and inflow control valves (ICVs). The ICDs are passive-control devices imposes additional pressure drop across the device with different mechanisms to make an inflow profile uniform along a horizontal well (Al-Khelaiwi and Davies 2007; Li et al. 2011a). The ICDs are deployed as a part of the completion base pipe (i.e., liner or casing) and packers with preset opening settings; their choking performance is not adjustable basically while recent technological advancements enable to make on/off settings of the ICDs after deployment (Li et al. 2011a; Dani 2024). The value of the ICDs has been studied for horizontal wells where inflow profiles are prone to non-uniform because of heel-toe effects (i.e., larger inflow from heel section caused by pressure drop along the lateral) and formation heterogeneity (Hill and Zhu 2008). The uniform inflow profile imposed by the ICDs potentially makes the well life longer by delaying selective gas/water breakthrough (Li et al. 2011a; Olsen et al. 2017). On the other hand, the ICVs are positive-control devices providing active control of reservoir inflow by adjusting openings of the valves from surface through hydraulic/electric control lines (as exemplified by Burton et al. 2019). The ICVs enable efficient reservoir management by optimal control in multilateral wells (e.g., Salamy et al. 2008), selective reservoir/layer diagnostics in multilayer reservoir (e.g., Pandey et al. 2015; Burton et al. 2019), and inflow control along the horizontal wells (e.g., Li et al. 2011b). It should be noted that, when a hydraulic-control system is employed for ICVs, each valve requires a dedicated control line, which leads to concerns in cost and long-term reliability.

Autonomous inflow control device (AICD) was introduced to the industry for regularization of gas/water production autonomously (i.e., without well interventions) as well as achievement of uniformly distributed inflow profiles (Mathiesen et al. 2011). The AICD is still a variant of the ICDs, though this device aimed at choking flow of low-viscous fluids autonomously with specially designed nozzles after the gas/water breakthrough, which could not be achieved with the conventional ICDs. The RCP (Rate Controlled Production) valve originally introduced by Mathiesen et al. (2011) uses difference of fluid's viscosity and density for its selective choking performance (i.e., difference in pressure-drop magnitude through the nozzle) with its specially designed disk-shaped valves. Its selective choking performance has been evaluated through a series of experiments under single-phase and multiphase conditions (Mathiesen et al. 2011; Halvorsen et al. 2016). The RCP-type AICDs have been piloted/deployed in a variety of fields in published literature including the North Sea (Mathiesen et al. 2011; Halvorsen et al. 2012; Halvorsen et al. 2016; Langaas et al. 2019; Langaas et al. 2020) and offshore Abu Dhabi (Ahmad et al. 2016; Agrawal et al. 2021). Halvorsen et al. (2016) demonstrated its field-wide implementation in the Troll field, North Sea for more than 30 wells with key findings that i) the oil production was significantly higher than simulated/expected, and ii) gas breakthrough occurs much later than expected. In addition, the fluidic-type AICDs have also been reported to be piloted/deployed in industry. This device utilizes specially designed flow channels inside the valve considering viscosity and density differences among the fluids without any moving parts to achieve the fluid-type identification and restriction of unwanted fluid for different viscosity ranges (Least et al. 2012; Least et al. 2013a; Corona et al. 2016; Corona et al. 2017a; Corona et al. 2017b). The fluidic-type AICDs have also been used in different regions including Ecuador (Least et al. 2013b), UAE

and Saudi Arabia (Iqbal et al. 2015; Corona et al. 2016). These autonomous regularization devices have been further evolved in industry, and pilots/deployments of new devices (e.g., autonomous inflow control valve, a variant of the AICDs (Taghavi et al. 2024)) are also reported in literature.

Performance of the downhole flow control devices has been formulated to reproduce experimental results. The pressure-drop across an ICD has been postulated to be proportional to squared flow rate (Lie et al. 2011a; Muradov et al. 2018; Langaas et al. 2019):

$$\Delta p_{ICD} = a_{ICD} q^2 = \frac{1}{2} \rho \left(\frac{q}{AC_d} \right)^2 \quad (1)$$

where a_{ICD} is called the ICD "strength" parameter, ρ is density of fluid, q is local volumetric flow rate, A is nozzle area, and C_d is the discharge coefficient. With prescribed nozzle size and average fluid density, calculated pressure drop as a function of flow rate can be matched against experimental results by adjusting the discharge coefficient. Langaas et al. (2019) suggested $C_d = 0.88$ using measured oil, water, gas properties with 2.0-mm-diameter ICD for the Alvheim reservoir. On the other hand, the pressure-drop across an RCP-type AICD has been proposed to follow the following relationship (Mathiesen et al. 2011; Voll et al. 2014; Moradi and Moldestad 2021):

$$\Delta p_{AICD} = a_{AICD} \left(\frac{\rho_{mix}^2}{\rho_{cal}} \right) \left(\frac{\mu_{cal}}{\mu_{mix}} \right)^y q^x \quad (2)$$

where a_{AICD} is called the AICD "strength" parameter, x and y are exponents for flow rate and viscosity, ρ_{cal} and μ_{cal} are the calibration density and viscosity (density and viscosity value at reference test conditions), and the ρ_{mix} and μ_{mix} are mixture density and mixture viscosity. These mixture properties are also defined as (Halvorsen et al. 2016):

$$\rho_{mix} = \alpha_{oil}^a \rho_{oil} + \alpha_{gas}^b \rho_{gas} + \alpha_{water}^c \rho_{water} \quad (3)$$

$$\mu_{mix} = \alpha_{oil}^d \mu_{oil} + \alpha_{gas}^e \mu_{gas} + \alpha_{water}^f \mu_{water} \quad (4)$$

where α is the phase volume fraction with a subscript denoting the fluid phase, and exponents from a to f are used for multiphase flow conditions. Compared to the ICD (Eq. 1), the AICD formulation is more complex and has three key parameters (a_{AICD} , x , and y), though, with appropriate regression method, its predictability has been justified against flow-loop experimental results (Mathiesen et al. 2011; Voll et al. 2014; Langaas et al. 2020; Moradi and Moldestad 2021). Its predictability under the multiphase flow condition, however, needs to be carefully evaluated because there is still room for improvement in the matching quality to the experimental results measured by varying water-cut and gas volume fraction (Halvorsen et al. 2016).

There are still technical challenges in application of the AICDs for its performance prediction. First, choice of the input parameters in the Eq. 2 is a key in practical applications for reliable prediction although not always are detailed experimental results available. Lack of experimental results with formation fluids leads to use of values provided by product vendors though its predictability becomes questionable. Table 1 summarizes set of AICD parameters in published literatures (Voll et al. 2014; Lei et al. 2020; Langaas et al. 2020; Agrawal et al. 2021; Moradi and Moldestad 2021). This highlights that (i) the strength parameter (a_{AICD}) may have relatively large variation from 2.660×10^{-7} to 8.07×10^{-4} in $\text{bar}/(\text{kg}/\text{m}^3)/(\text{rm}^3/\text{day})^x$ (this value should be treated with caution because a_{AICD}/ρ_{cal} is the proportional constant in Eq. 2; Moradi and Moldestad (2021) postulated 10 times larger value of ρ_{cal} , the apparent a_{AICD} seems also 10 times larger than others), (ii) the flow exponent (x) varies from 2.15 to 3.52, and (iii) the viscosity exponent (y) seems to vary from 0.11 to 1.0 (mode is close to 0.6). While Halvorsen et al. (2012) stated the typical value of x varies between 3.0 and 4.0, the laboratory-based estimation in these references revealed the flow exponent seems less than the range. Second, in-situ performance of the AICD nozzles at downhole conditions is uncertain (i.e., consistency of the nozzle choking performance between surface experimental results and downhole

diagnostic results) because there are limited number of publications discuss the in-situ nozzle performance. Langaas et al (2020) presented a diagnostics with the pressure transient analysis (PTA) for a wells with the (A)ICDs indicating potential rate-dependent nature of the skin factor and a history-matching outcome (i.e., needs of more choking in AICDs and less in ICDs) of chemical PLT and downhole flowing pressure, which justifies importance in calibration of the in-situ AICD parameters for improved forecasting. Though the values to be used for these key inputs are different for each field, more field examples and practical methods for the AICD calibration using measured dataset are valuable for future applications of this technology.

Table 1—Summary of AICD parameters in published literature ("N/A" denotes "not available").
 Langaas et al. (2020) presented parameters of 5 mm and 7.5 mm AICDs (upper and lower column, respectively) though their units were in bar-day/kg (supposing $x = 1$ in this work).

Literature	Parameter										
	a_{AICD} bar/(kg/ m ³)/(rm ³ / day) ^x	x -	y -	ρ_{cal} kg/m ³	μ_{cal} cP	a -	b -	c -	d -	e -	f -
Voll et al. (2014)	4.004×10^{-5}	2.559	0.694	1,000	1	1	1	1	1	1	1
Voll et al. (2014)	3.554×10^{-5}	2.68	0.592	1,000	1	1	1	1	1	1	1
Lei et al. (2020)	1.500×10^{-5}	2.5	0.6	1,000	0.5	1	N/A	1	1	N/A	1
Langaas et al. (2020) – 5 mm	1.380×10^{-6}	3.52	0.608	1,000	1	1	1	1	3	2	0.5
Langaas et al. (2020) – 7.5 mm	2.660×10^{-7}	3.52	0.608	1,000	1	1	1	1	3	2	0.5
Moradi and Moldestad (2021)	8.077×10^{-4}	2.155	0.110	10,000	1	1	N/A	1	1	N/A	1
Agrawal et al. (2021)	1.300×10^{-5}	2.7	1.0	750	0.79	1	1	1	1	1	1

This study focuses on a giant carbonate reservoir located offshore Abu Dhabi. The target field consists of several reservoirs with oil production. After the discovery, this field has been developed with dump-flood water injection, peripheral water injection, and crestal gas injection, and currently the reservoir pressure has been maintained by the combination of the powered water injection from the down-flank side and gas injection in the structural crest (Montani et al. 2024; Ito and Hiraiwa 2025). Reservoir-A is one of the main reservoirs, and the gas cap expansion near the crest area and remaining water risks in the mid-dip area caused by the dump-flooded water are simultaneously occurring (Figure 1). To increase oil production rate and sustain it for a long duration, additional development plan is in progress and new infill drilling of long horizontal wells is a part of this. However, in some regions, the new infill drilling faces challenges to encounter expanded gas zones and to increase production gas-oil-ratio (GOR) which potentially decrease the gas-injection efficiency for pressure maintenance. The AICD has been considered as one of the effective methods to control gas production among other potential countermeasures (e.g., careful well location and layer selection, gas migration control by inner ring water injection fence in Koeck et al. 2019, and chemical/mechanical gas shut-off technology). Hence, the AICD has been deployed in several wells of the Reservoir-

A to evaluate its capability for, in particular, gas rate regularization under the in-situ fluid characteristics and formation properties of the subject reservoir.

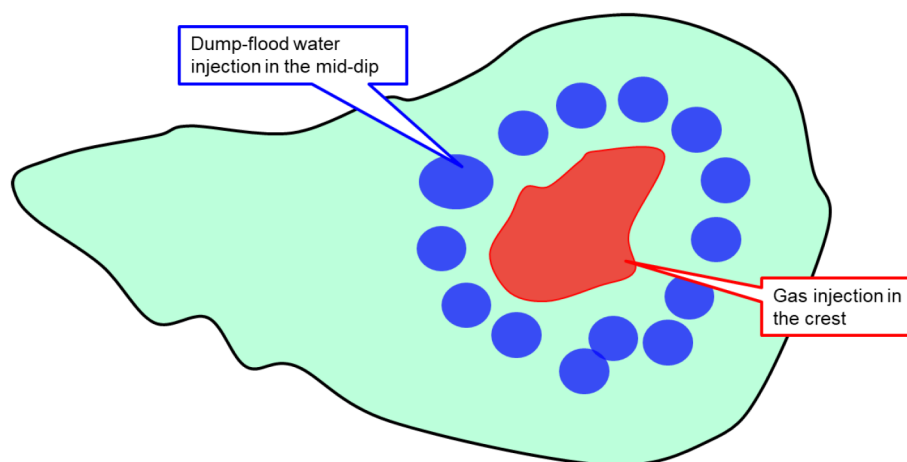


Figure 1—Conceptual fluid distribution map in the subject field. Color denotes fluid type conceptually; green denotes oil zone, blue denotes water zone and red denotes gas zone.

The objective of this paper is to demonstrate ranges of the AICD parameters through well performance diagnostics on the AICDs piloted/deployed for gas production regularization in the giant carbonate field located offshore Abu Dhabi. The well performance diagnostics targets three wells completed with the AICDs as the field pilot/deployment of this technology. It tries to calibrate the AICD parameter ranges through history-matching of the actual inflow profiles and simplified 1D wellbore simulation model. This work also presents several case studies using reservoir simulation models and an advanced wellbore modeling capability (i.e., multi-segment well model) within the calibrated ranges of the AICD parameters in the subject field for improved understanding of the performance of wells equipped with AICDs. Readers may obtain improved understanding on differences in the designed and actual AICD parameter range through in-depth analysis of the dataset and expected long-term behavior of the well equipped with the AICDs under different scenarios, serving as a reference for the future usage of this technology in analog reservoirs.

Method

An overall study workflow is presented in Figure 2, which consists of review/analysis of performance of the wells completed with AICDs, AICD parameter calibration, and forecasting of AICD performances for future infill planning in the subject field.

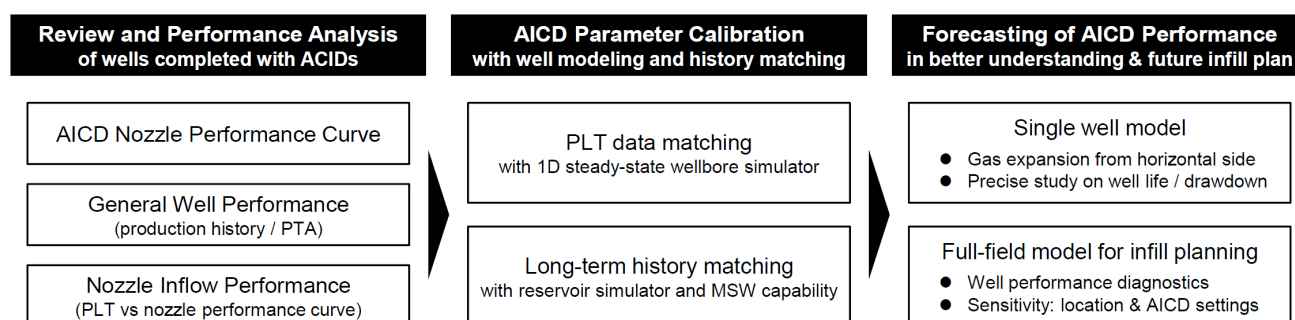


Figure 2—Overall study workflow (PTA, PLT, MSW denote pressure transient analysis, production logging tool and multisegment well).

The performance review and analysis present a series of data review and analyses of the three pilot wells completed with the RCP-type AICDs in the subject field. This section intends to demonstrate (i) anticipated performance of the AICD nozzles under the fluid/formation characteristics, (ii) their impacts in well production history, and (iii) comparison of actual performance against the designed one to evaluate magnitude of potential deviation from design and reliability of the device for future installation. The RCP-type AICD nozzle performance (i.e., pressure drop vs flow rate) is expected to follow the proposed relationship Eq. 2, so appropriate input data is used to draw the performance curve for improved understanding of the expected additional pressure drop in the field. Subsequently, the impacts of the additionally induced pressure drop are investigated through the review of production history and pressure build-up test interpretation. Finally, the inflow profiles estimated by the PLT survey are used to estimate inflow rate per nozzle with consideration of reservoir pressure and wellbore flowing pressure, which enables us to infer any potential nozzle leakage.

The AICD parameter calibration tries to diagnose appropriate range of the AICD parameters through history-matching with wellbore modeling, which serves as a key input for the AICD modeling in the reservoir simulation models for the long-term well performance evaluation. This section focuses on one of the three wells with consideration of data availability. The parameter calibration includes PLT data matching using i) a 1D steady-state numerical commercial wellbore simulator and ii) long-term well performance matching using a numerical reservoir simulator. In the matching process, different scenarios are tested through one-variable-at-time sensitivity study considering the three AICD parameters, and possible scenarios are ranked based on the history-matching quality. It should be noted that the reservoir simulation study utilizes the MSW capability and the nozzle performance curves to incorporate impact by the AICD nozzles precisely.

The last part presents a series of forecasting results under different scenarios to demonstrate expected performance provided by the AICDs in the subject field. The forecasts given by the reservoir simulation model seem improved and more confident by using the AICD parameters calibrated against measured dataset. At first, a single well model is constructed to predict the AICD performance under the hypothetical horizontal gas breakthrough scenario with refined grid cells, which assesses impacts on drawdown with imposed additional pressure drop, in-depth fluid movement after the gas breakthrough, risks of nozzle leakage, and potential oil gain compared to no-AICD scenario. Finally, these findings are incorporated into infill planning using the full-field model considering the impact of the gas coring. The value of the AICDs in future wells is deeply investigated under different scenarios.

Results and Discussion

Review and Performance Analysis of Wells Completed with AICDs

Description of the Pilot Wells with AICDs. This work focuses on three (3) wells completed with the AICDs as a field-level pilot/deployment program in Reservoir-A (Figure 3). All these wells have a 6" horizontal openhole drain with 2,000-3,000 ft in length and completed with 4-1/2" liner, multiple swellable packers and 16-20 sets of the RCP-type AICDs. Chronologically, the AICD well-3 was completed at first with shortest drain length (2,000 ft) and smaller number of AICDs (i.e., 16 AICDs), and, subsequently, remaining two wells were drilled/completed with longer drain length and increased number of the AICDs to improve well productivity. All these wells were completed with same vendor's RCP-type AICDs. Each AICD has three sockets in the base pipe, and two of them are equipped with the production valves (PVs) and the last one with the stimulation valve or blank valve. It should be noted that the AICD well-3 was perforated on the 4-1/2" liner body at three perforation intervals later to improve its productivity.

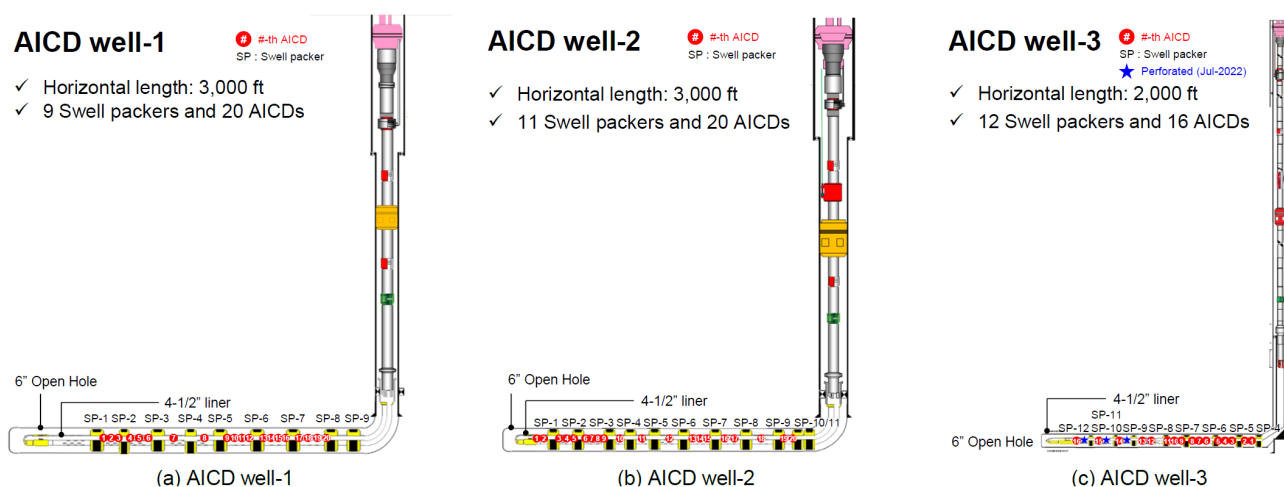


Figure 3—Well completion schematics of the three AICD wells.

AICD Nozzle Performance Curve. The RCP-type AICDs deployed on these three wells are expected to follow the selective choking relationship of Eq. 2 with multiphase constitutive relations (Eq. 3 and Eq. 4). The user input parameters of the device used in the subject reservoir are summarized in Table 2 where ρ_{cal} and μ_{cal} are calibration values specific to each device and all the exponents used in Eq. 3 and Eq. 4 are assumed to be 1.0 in the current work for simplicity. Designed AICD nozzle performance curves using Eqs. 2-4 and inputs in Table 2 are shown in Figure 4. It should be noted that the pressure-drop curves of oil and gas seem overlapped using the fluid properties in the subject reservoir although published cases (e.g., Halvorsen et al. 2016; Langaas et al. 2020) show larger difference between oil and gas curves, which will be further discussed later.

Table 2—Summary of AICD parameter and device specific value.

Type of Input	AICD Parameter (Unit)	Value
User input parameter	a_{AICD} (bar/(kg/m ³)/(rm ³ /day) ^x)	1.3×10^{-5}
	x (-)	2.7
	y (-)	1.0
Device specific value	ρ_{cal} (kg/m ³)	750.72
	μ_{cal} (cP)	0.79

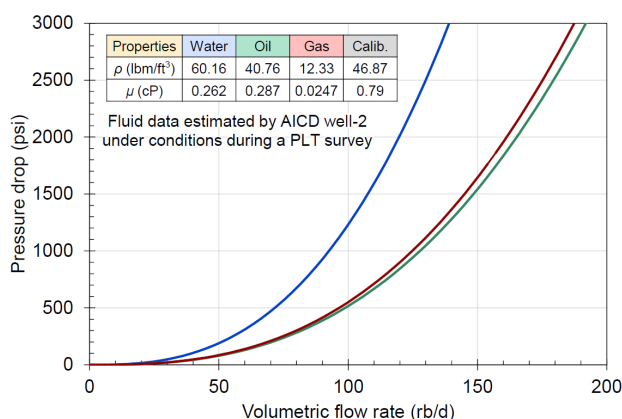


Figure 4—Designed AICD nozzle performance curve for a single AICD nozzle (drawn with Eq. 2 and input data in Table 2).

General Well Performance Review. Figure 5 summarizes production history of the three wells covering oil production rate, GOR, water cut, and wellhead flowing pressure. This figure also contains sequences of events such as PLT surveys, noticeable water/gas production trend changes, and additional perforation jobs in AICD well-3. The AICD well-1 showed the gradual decrease in oil production rate and wellhead flowing pressure. Water breakthrough was observed in Year 2-Aug, though GOR trend was stable. PLT surveys were conducted twice in Year 2-Feb (just after the production start) and Year 3-Jan (after water cut increased). The AICD well-2 showed decline in oil rate with slight variation in GOR by at most 100% difference (i.e., maximum GOR value is less than twice of the original GOR level). PLT survey was conducted only once for this well in Year 2-Feb (just after the production commencement). The AICD well-3 showed relatively stable oil rate but suffered from low wellhead flowing pressure. The first PLT survey was conducted after the water cut increased. Subsequently, additional perforation jobs were conducted at in total three intervals on the 4-1/2" liner body to reduce completion pressure drop by making the flow bypass the AICDs. Its benefit was clearly shown in instantaneous increase of the wellhead flowing pressure indicating improvement of well productivity, but, in 3 months of production, gradual GOR increase occurred subsequently. Second PLT survey was conducted in this well in Year 2-Jul, which was after the additional perforation job.

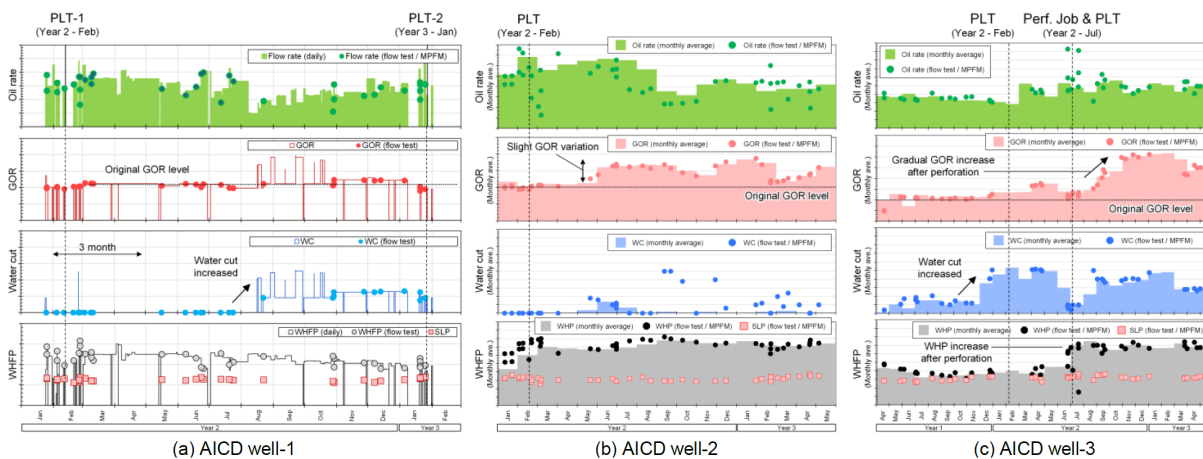


Figure 5—Summary of production history and sequence of events of the 3 wells completed with the AICDs.
In these plots, WC(T) denotes water cut, WH(F)P denotes wellhead (flowing) pressure, MPFM denotes multiphase flow meter, and SLP denotes sea-line pressure. Y-axis of these plots are not in the same scale.

Table 3 shows the interpretation of pressure transient analysis (PTA) covering the three AICD wells and wells with different completion type (i.e., ICD and vertical well with perforation). The interpreted skin factor at the wells with AICDs and ICDs shows relatively larger value than vertical well. However, this result is not a direct outcome of AICDs because not only the contribution of completion types but also formation damage and acid stimulation effectiveness will have an impact on skin factor.

Table 3—Summary of pressure transient analysis interpretation for the AICD wells with offset reference well data of an ICD completion well and a vertical well (completed with cased/cemented and perforation).

Item	AICD well-1	AICD well-2	AICD well-3	[Offset reference] ICD well	[Offset reference] Vertical well
Boundary condition	Fault	Fault	Infinite acting	N/A	Infinite acting
Flow capacity (mD-ft)	3,887	2,261	5,275	N/A	7,091
Skin factor (-)	18.17	0.145	0.315	N/A	4.85
Remarks	-	Uncertainty in interpretation	Difficult to interpret (data quality)	-	-

AICD Nozzle Inflow Performance. In-depth evaluation of the AICD nozzle performance starts from review of an inflow profile during the 1st PLT survey. Figure 6 shows average inflow rates from each AICD PV estimated with the 1st PLT survey. The PLT interpretation suggests inflow rate from each AICD up to the hold-up depth of each well, and the inflow from each PV is simply assumed by dividing the inflow rate by 2 because 2 PVs are set in each device. At the time of the surveys, no significant gas/water production was observed at surface. The nozzle inflow rate summary indicates some AICD PVs potentially accommodate a higher inflow rate than 150 rb/d/nozzle. The AICD nozzle performance curve (Figure 4) estimates the pressure drop to be more than 1,500 psi, though, with consideration of much lower drawdown measured during the PLT survey, the choking performance of 2-4 nozzles (in all these AICD wells) seem lower than designed, inferring potential variation of AICD nozzle performance and/or, in one of the extreme scenarios, possibility of device malfunction and nozzle leakage.

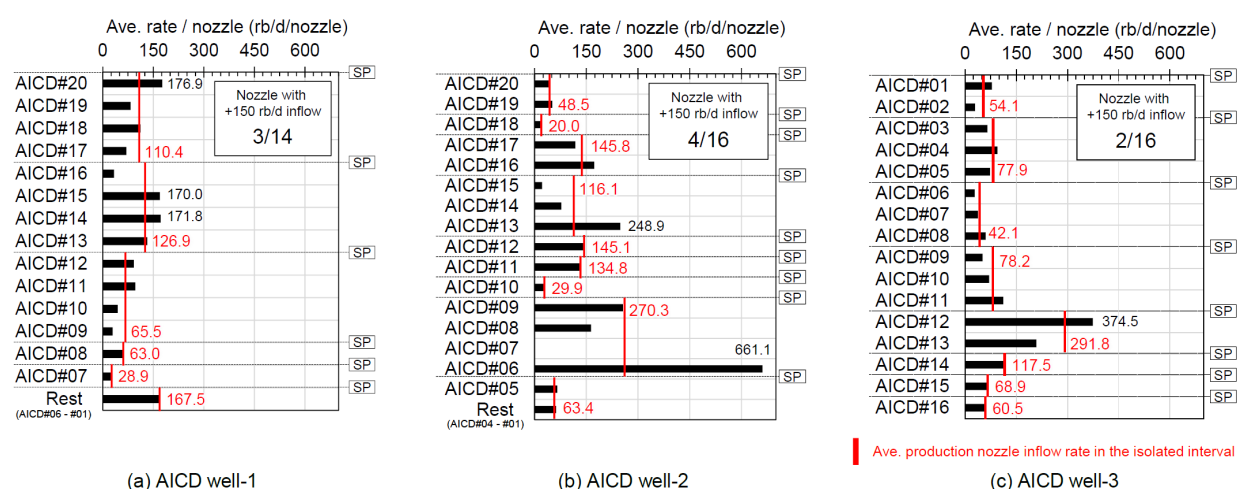


Figure 6—Summary of inflow from each AICD production valve estimated with the 1st PLT survey results of each well. Black bar denotes measured inflow rate from each nozzle and red line denotes average nozzle inflow rate within the same interval isolated by swell packers.

Time-lapsed PLT surveys conducted in the AICD well-1 and AICD well-3 provides temporal change in inflow profiles associated with changes in production condition. Figure 7 compares the AICD nozzle inflow profiles during the 1st and 2nd PLTs in the AICD well-1. The 1st PLT in Year 2-Feb shows relatively uniform inflow profile, though the 2nd PLT in Year 3-Jan seems extremely large inflow (e.g., more than 300 rb/d/nozzle) from some AICDs such as #02 and #19. This suggests way higher pressure drop in these nozzles compared to the 1st PLT, but, again, this extremely high drawdown was not observed in the 2nd PLT survey. This drastic change in inflow profile indicated that AICD performance is possibly altered during the continuous production in a year. It should be also noted that this interpretation depends on the quality/uncertainty in the PLT data/analysis itself; PLT surveys in a horizontal well under multiphase condition typically use advanced multi-sensor tools including spinners and capacitance/resistivity/optical tools where interpretation complexity leads to increased uncertainty (as exemplified by Liao et al. 2013). Compared to the AICD well-1, the AICD well-3 experienced the additional perforation job during the production period making the flow condition different. Figure 8 compares the inflow profile before and after additional perforation job. The newly added production intervals by perforation jobs (i.e., near #14, #15 and #16) show significantly higher inflow rate (3-6 times larger). This result indicates that the AICDs surely impose additional pressure loss and try to regulate inflow profile into the completion/well.

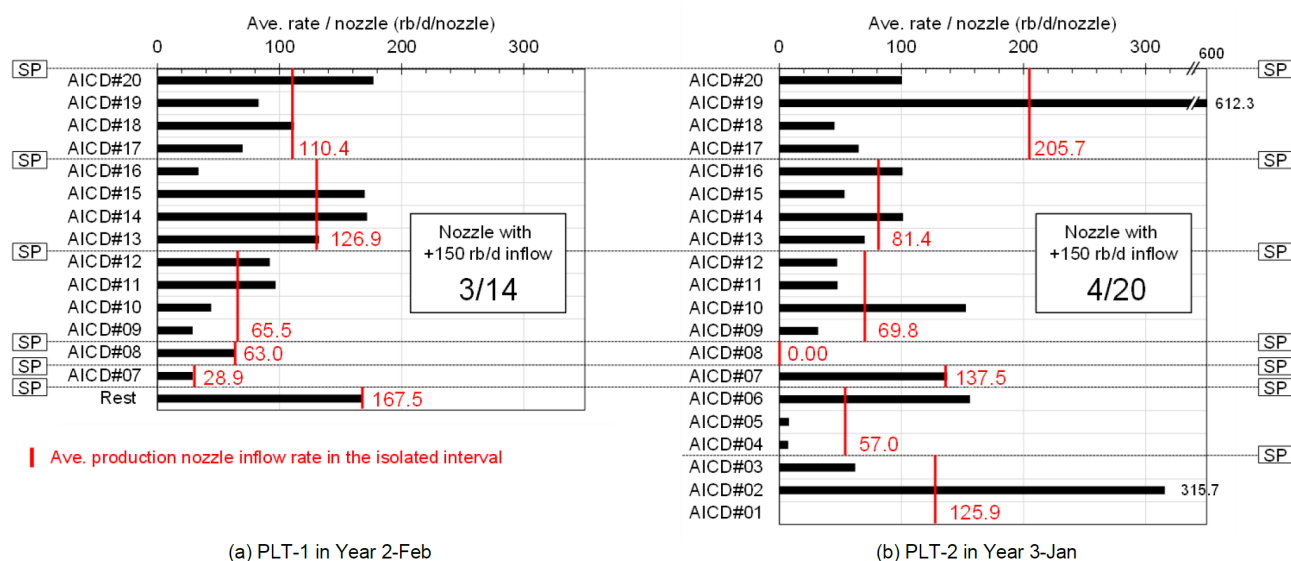


Figure 7—Comparison of the nozzle inflow profile in the AICD well-1 within 1 year production.

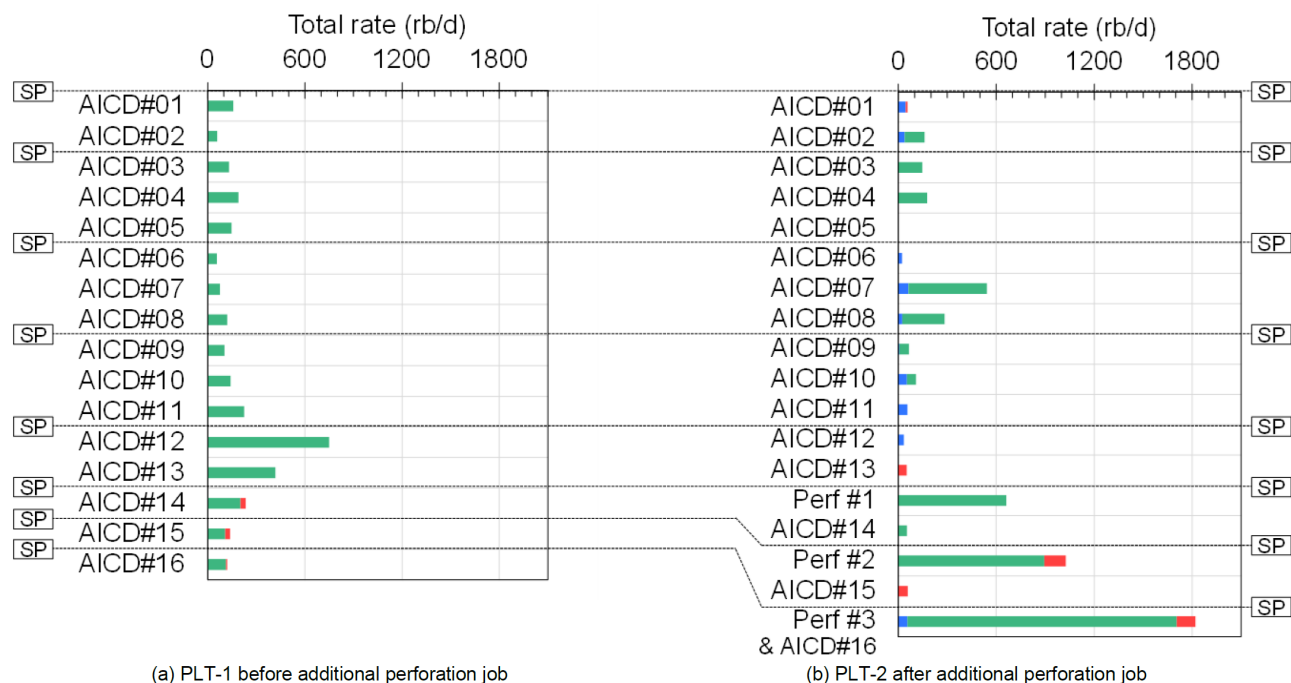


Figure 8—PLT inflow profile comparison at AICD well-3 before and after additional perforation job. Color denotes different fluid type; green denotes oil inflow, blue denotes water inflow, and red denotes gas inflow.

The outcome of performance data review on the AICD wells in Reservoir-A covering long-term production, PTA and nozzle inflow performance with the PLT surveys is summarized as follows:

- AICDs surely impose additional pressure drop as identified in the inflow comparison of the AICD well-3 with extremely higher inflow rate from the perforated intervals after the additional perforation jobs.
- Interpreted skin factor by the PTA seems higher in the AICD wells compared to vertical well, though it is not conclusive because other potential reasons (e.g., formation damage and stimulation effectiveness) seem to remain as the cause of variation of the skin.

- With the given PLT results, 1-2 AICDs may potentially show lower pressure drop compared to the estimation by the designed AICD performance curve. This indicates possibility of variation in AICD nozzle performance and/or device malfunction or nozzle leakage.
- The AICD nozzle performance seems to vary with time during continuous production.

AICD Parameter Calibration with Well Modeling and History Matching

PLT Data Matching with 1D Steady-state Wellbore Simulator. The 1D model construction targets the AICD well-2 with consideration of data availability incorporating its well trajectory, completion diagram, static and dynamic reservoir properties based on logging data and a reservoir simulation model. The initial model was updated to reproduce estimated static pressure and flow capacity estimated by the PTA through adjustment of permeability profile uniformly along the wellbore. With use of the updated 1D model, history matching of the PLT data (i.e., inflow and pressure profile) is tried by uniformly adjusting the AICD parameters (a_{AICD} , x and y) for all the AICDs taking an approach of OVAT (One-Variable-At-a-Time). In addition, the potential nozzle leakage scenario was also investigated by assuming negligible pressure drop at some AICDs to reproduce possibility of the higher inflow rate than 150 rb/d/nozzle. Table 4 shows the AICD parameter setting for five (5) cases.

Table 4—Summary of AICD parameter adjustment for the Well-2 PLT matching.

Case	AICD Parameter (Unit)		
	a_{AICD} (bar/(kg/m ³)/(rm ³ /day) ^x)	x (-)	y (-)
Original design	1.3×10^{-5}	2.7	1.0
a_{AICD} adjustment	1.1×10^{-5}	2.7	1.0
x adjustment	1.3×10^{-5}	2.7	1.0
y adjustment	1.3×10^{-5}	2.7	1.0
Leakage	[Working] 1.3×10^{-5} /[Leakage] 1.0×10^{-9}	[Working] 2.7 / [Leakage] 2.7	[Working] 1.0 / [Leakage] 1.0

Results of the history matching with the above five cases against the observed PLT data are shown in Figure 9. All the sensitivity cases except for the ‘y’ adjustment case demonstrates improved quality in the matching against design parameter case; the rest of the 3 cases shows excellent matching quality in pressure profile against the original case and reasonable matching quality in the total downhole flow rate (equivalent level with the original case). This outcome indicates existence of multiple combination of the AICD parameters to reproduce the actual PLT performance, and further calibration of these parameters seems required for reliable performance forecasting for long-term AICD performance. It should be noted that this work employed the OVAT approach but in particular the value range of the ‘y’ exponent is higher uncertainty because this well did not experience the gas/water breakthrough at the time of the PLT survey.

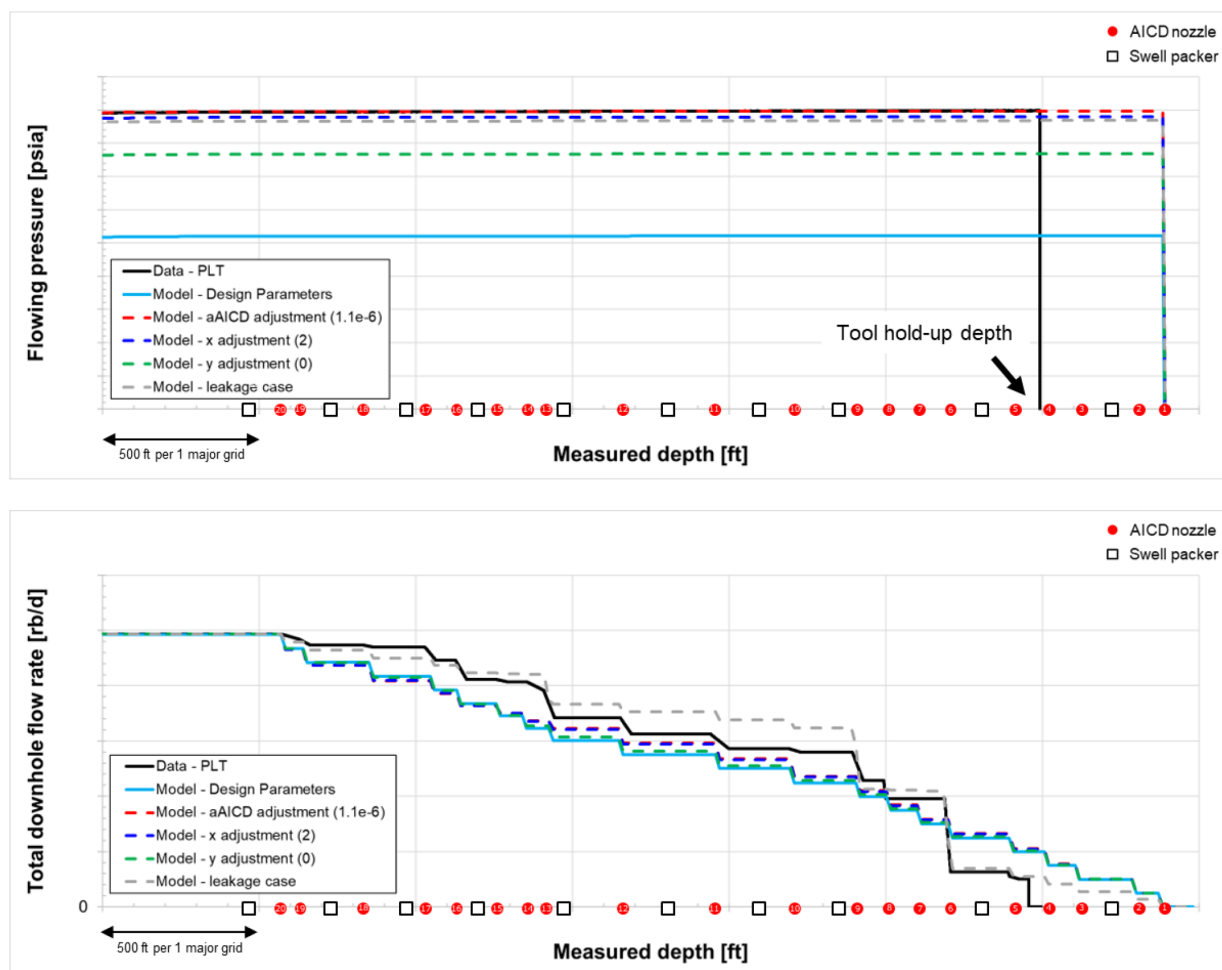


Figure 9—Results of history matching against (top) measured pressure profile and (bottom) profile of total downhole rate along the AICD well-2. The tool hold-up depth was near AICD #4 and no measurement data exists beyond this depth caused by inaccessibility of the PLT tool.

Long-term Pressure History-matching by Numerical Reservoir Simulator. This section tries to further calibrate the AICD parameters accounting for the long-term production history of the AICD well-2 through history-matching with a single well reservoir simulation model. The single well reservoir simulation model is defined as 2,000 ft $\times$ 4,000 ft in area and 40 ft in height, and its rock and fluid properties are referred to openhole logging data and an existing full-field reservoir simulation model. To precisely reproduce pressure-drop across the AICDs, the well was modelled with the MSW option in a reservoir simulator which discretizes well flow-path (i.e., liner, AICD, annulus) into separate segments and compute flow dynamics in the wellbore.

The history matching starts from reproduction of 1+ year production history of the AICD well-2 and downhole pressure data (i.e., in particular shut-in pressure data) during the pressure gauge was deployed in the well by adjusting pore-volume multiplier. Once it was confirmed that the reservoir has sufficient energy to reproduce its production history and reasonable level of matching in the shut-in pressure, the above 3 possible scenarios on the AICD parameters (i.e., ' a_{AICD} ' adjustment, ' x ' adjustment and potential leakage scenario) are tested to check if they can reproduce the flowing bottomhole pressure data. Figure 10 shows results of the history-matching on oil production rate and bottomhole pressure among 3 cases. All the cases successfully demonstrate reproduction of the daily oil production rate. With respect to the bottomhole pressure, though all 3 cases present reasonable matching during the PLT survey in Year 2-Feb, a_{AICD} adjustment case shows a larger deviation in pressure at the beginning of Year 2-Mar at higher production rate and ' x ' adjustment scenario and potential leakage scenario show better matching quality. This indicates that

these two scenarios can reproduce the well performance under wider production conditions; these scenarios are used for the subsequent performance forecasting.

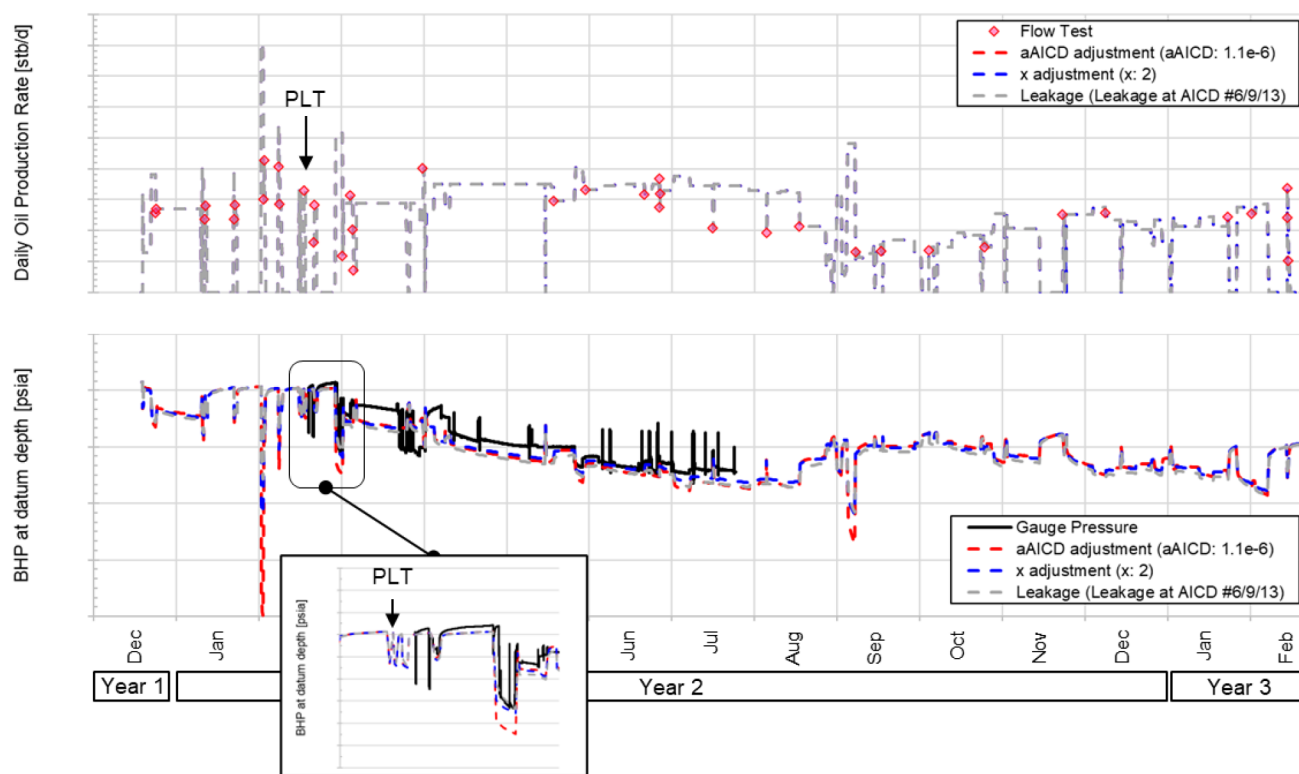


Figure 10—History-matching results of the AICD well-2 for long-term production using the single well reservoir simulation model. BHP denotes bottomhole flowing pressure.

Forecasting of AICD Performance in Better Understanding and Future Infill Plan

The integrated approach of the AICD parameter calibration successfully demonstrates potential combinations of the AICD parameters to reproduce actual well performances. The calibrated AICD parameter case with the volumetric flow rate exponent (x) and the leakage case are evaluated for long term performance evaluation in this section for assessing the AICD performance in forecasting.

AICD Well Performance Evaluation by a Single Well Reservoir Simulation Model. With the calibrated AICD parameter sets, this section tries to predict the AICD performance under the hypothetical horizontal gas breakthrough scenario with refined grid cells. This tries to provide insights in assessment of the impacts on drawdown with imposed additional pressure drop, in-depth fluid movement after the gas breakthrough, risks of nozzle leakage, and potential oil gain compared to no-AICD scenario. This hypothetical scenario uses same settings with the one presented in the single well reservoir simulation model through the gas breakthrough is assumed potentially from the heel section with consideration of actual location in the reservoir and cap gas expansion.

Figure 11 shows performances of the AICD well-2 under the hypothetical gas flooding scenario for the "No AICD case", "AICD case" and "Leakage case". The "No AICD case" assumes no completion pressure drops at the AICD location in the MSW, while the "AICD case" and "Leakage case" assume completion pressure drop with AICD parameter settings of the ' x ' adjustment case and the potential leakage case, respectively. For all the cases, well will reach to GOR threshold in the current simulation assumption, while installation of the AICDs potentially extends the well life by 113 days with the mitigation of GOR increase. The cumulative oil production increases by 30% from the No AICD case. The AICD case also increases drawdown by 6-10 time compared to the No AICD case, and the impact on drawdown needs to be

carefully investigated to apply this technology in low reservoir pressure condition and at low productivity wells. Leakage case diminishes the AICD effectiveness on well life extension and accelerates the gas breakthrough compared with No AICD case. The predicted drawdown is lower than the AICD case caused by the negligible friction loss at the some AICDs even with higher inflow rate, while it still extends well life by 43 days compared to the No AICD case.

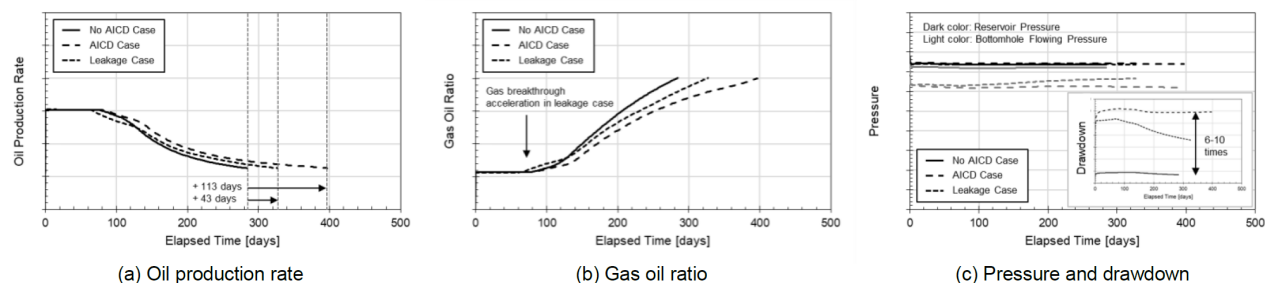


Figure 11—AICD Well-2 performance under the hypothetical gas flooding scenario.

Figure 12 shows gas saturation map of all cases and corresponding gas saturation difference map between AICD and No AICD case and between Leakage and No AICD cases (2 months after production start). AICD case mitigates the spiky gas front in the middle of horizontal section with highest permeability along the wellbore. At the toe section with relatively high permeability, the gas front shows slow-down by AICD application. Leakage case accelerates the spiky gas front near the leaking AICD and causes accelerated gas breakthrough. Like the AICD case, gas front shows slow-down at the toe section, while gas flooding acceleration is clear relationship with the location of leaking AICDs. The leaking AICD will become a potential risk for early gas breakthrough.

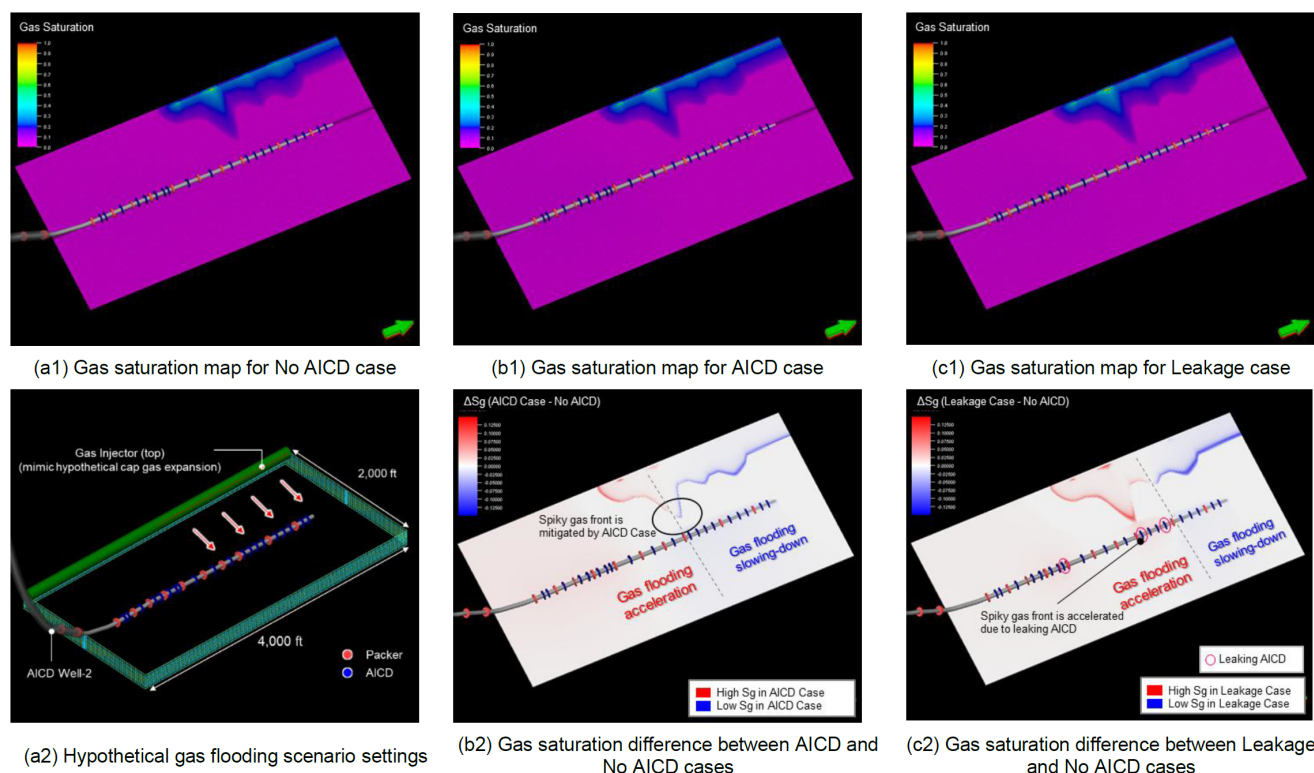


Figure 12—Gas saturation maps and its difference map between the AICD/Leakage case and the No AICD case in 2 months after production start under the hypothetical gas flooding scenario.

AICD Well Performance Evaluation in Full Field Simulation.

AICD well settings

For the evaluation of AICD performance at future well, target well location is selected with consideration of gas and water distribution. The target well (Well-A) is to be drilled in the northern area and have a horizontal section of approximately 8,200 ft to recover the oil from the wide area. Well-A is supposed to face challenges of water/gas breakthrough caused by its location between water distribution under the historical water injection and crestal gas expansion. The AICD well completion consists of 40 AICDs with 20 isolated sections uniformly divided by swell packers. This is intended to build uniform inflow distribution for the entire horizontal section. AICD nozzle performance in the numerical simulator is based on the volumetric flow rate exponent (x) adjustment scenario. In addition to the AICD scenario, conventional completion case with pre-perforated liner (No AICD Case) was also evaluated as baseline performance. Figure 13 shows the Well-A location in Reservoir-A and a well cross-sectional map with AICD completion.

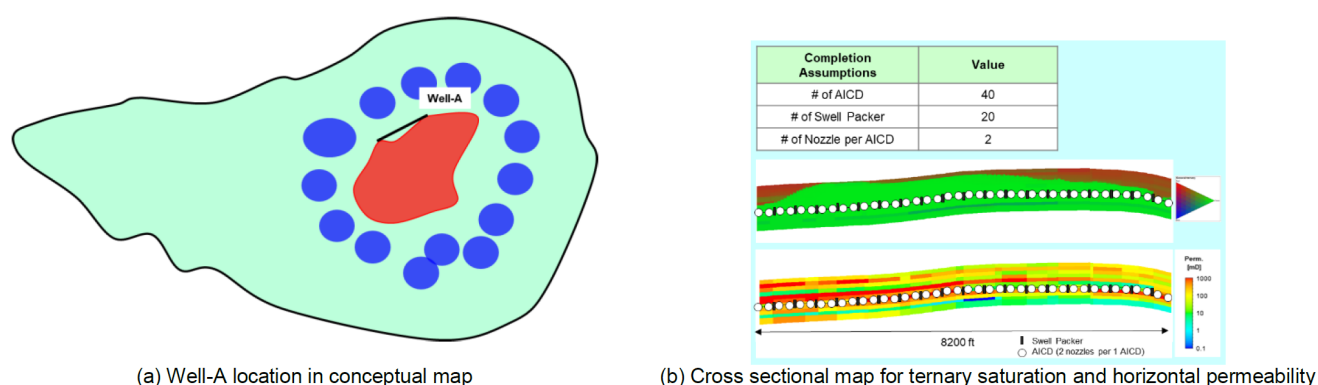


Figure 13—Location and completion summary of Well-A for performance evaluation in the full-field reservoir simulation model.

The well production performance is controlled primarily by oil rate and to account for the AICD performance, constraints of minimum bottomhole flowing pressure, minimum wellhead flowing pressures, maximum GOR are also considered in this work. These minimum bottomhole flowing pressure and wellhead flowing pressure are decided with consideration of the margin from saturation pressure and sealine pressure respectively. The maximum GOR is applied to avoid injection gas cycling in the reservoir and manage the excessive gas expansion in a certain direction.

AICD well performance evaluation

AICD effectiveness at Well-A is evaluated based on the well performance comparison between No AICD Case and AICD Case (Figure 14). For both cases, well experiences gradual increase in GOR after production start and stops production due to maximum GOR constraint. AICD Case shows the mitigation of GOR increase trend and well life extension by 2 months compared with No AICD Case. Resultantly, cumulative oil production increases by 30% of oil production in No AICD Case. Reduction in bottomhole flowing pressure is also predicted because of the increase in pressure drop via device as an unavoidable impact by the AICD implementation. Drawdown in AICD Case increases by 10 times compared with No AICD Case under high GOR condition just before the well shut-in.

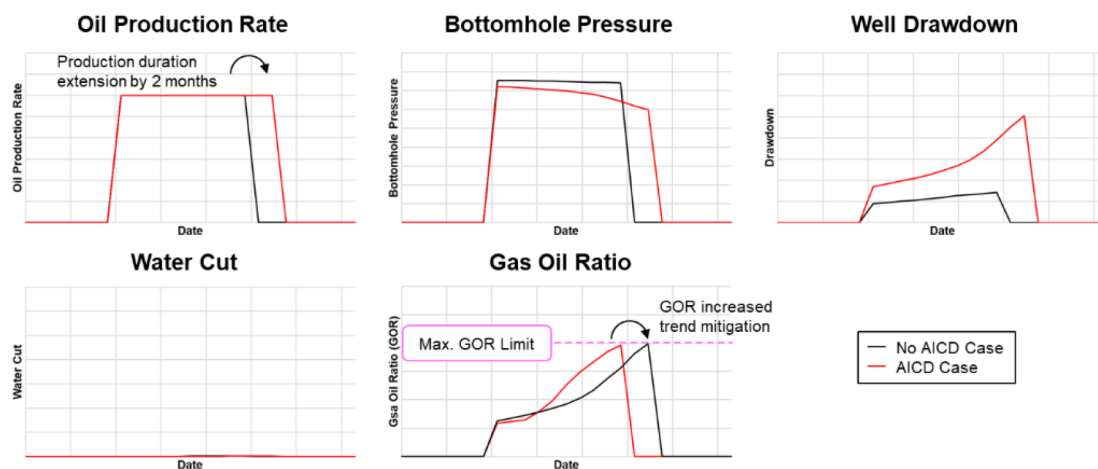


Figure 14—Well-A production improvement by AICD in comparison with No AICD case.

AICD has two advantages to improve the well performance. One is the building uniform inflow distribution by imposing additional pressure drop related to volumetric inflow rate. This will mitigate sharp entry from the high permeability sections and enhance oil recovery from the low permeability ones. Sectional inflow and GOR distribution comparison between No AICD Case and AICD Case is shown in Figure 15. More than 60% of inflow comes from the high permeability sections in the middle of horizontal drain for No AICD Case at the initial condition. The production contribution from the latter half near the toe section is significantly small. At the end of production, GOR increases mainly in the high permeability section with higher production contribution, while GOR is stable at the initial level in the low permeability section. It indicates that the oil in low permeability section is not effectively recovered in No AICD Case. AICD Case shows uniform inflow distribution in the entire horizontal section at the initial condition. Uniform GOR increase in the entire well section at the end of production shows the ideal gas flooding front creation in AICD Case.

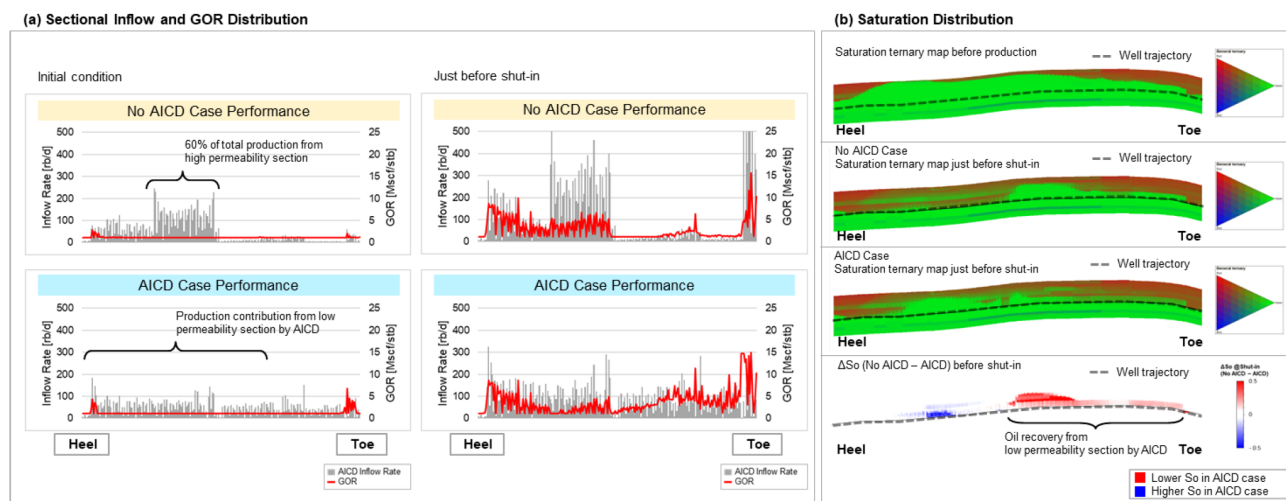


Figure 15—Sectional inflow and GOR distribution comparison at the initial and production stop.

The other advantage is the selective regulation of unwilling fluid. AICD nozzle performance curve (Eq. 2) is a function of volumetric flow rate and fluid properties including viscosity and density. For the evaluation of selective regulation performance, the term related to fluid properties in AICD nozzle performance equation (Eq. 5) is considered here which is defined as β in this work:

$$\beta = \left(\frac{\rho_{mix}^2}{\rho_{cal}} \right) \left(\frac{\mu_{cal}}{\mu_{mix}} \right)^y \quad (5)$$

The fluid properties at the AICD nozzle are extracted from the simulation results and summarized in Figure 16. Oil and gas properties are based on the simulation input as reference. Both fluid density and viscosity monotonically decrease with GOR increase due to the mixture of oil and gas. On the other hand, β shows a concave shape with a minimum value at 2–4 Mscf/stb of GOR. Even for the fluid with 8 Mscf/stb of GOR, β is still lower than oil. This plot indicates that friction loss for mixture fluid with GOR of 8 Mscf/stb is less than that of oil under the same reservoir volumetric flow rate. This result elucidates that AICD will not regulate gas production but accelerate high GOR fluid production under certain fluid properties and AICD parameter y .

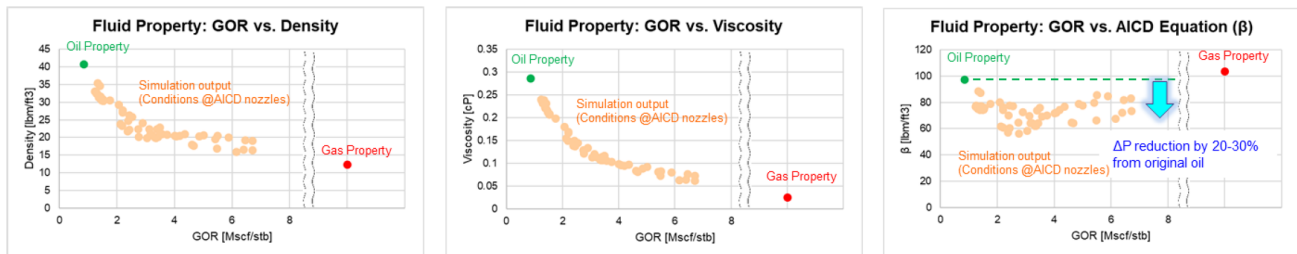


Figure 16—Relationship between fluid properties and β with GOR in the simulation result.

In the reservoir simulation, AICD well shows the mitigation of GOR evolution trend and the increase in cumulative oil production, while the detailed performance review revealed the effectiveness of two (2) advantages as below:

- Uniform inflow distribution can be achieved by AICDs with improvement of oil production in the low permeability zone. At the end of well life, GOR also increases uniformly in the entire well, which indicates an ideal gas flooding achieved by AICDs.
- Selective gas regulation cannot be expected for fluids properties in Reservoir-A. The calculated pressure drop for relatively high GOR fluid can be less than oil. The effectiveness of selective regulation will be subject to fluid properties and parameter y ,

Sensitivity study for AICD well location

During the drilling preparation of production wells, well location selection is an important process for well performance optimization with consideration of several risks. In Reservoir-A, the crestal gas distribution is one of the major concerns and needs to be taken into consideration for well location selection. AICDs are expected to mitigate the early gas breakthrough by building an ideal gas flooding condition, while this impact can be different subject to each well location. For the evaluation of AICD effectiveness from the viewpoint of well location, two (2) well locations were prepared for Well-A in addition to Base Location as shown in Figure 17. Location-1 is shifted toward the down-dip side by 700 ft from Base Location. Before production starts, there is negligible gas distribution above the well compared with Base Location. Location-2 is shifted toward the up-dip side by 700 ft, where massive gas is distributed above the well. For the evaluation of AICD effectiveness, the well length, number of AICDs and packers and AICD parameters are the consistent for all locations.

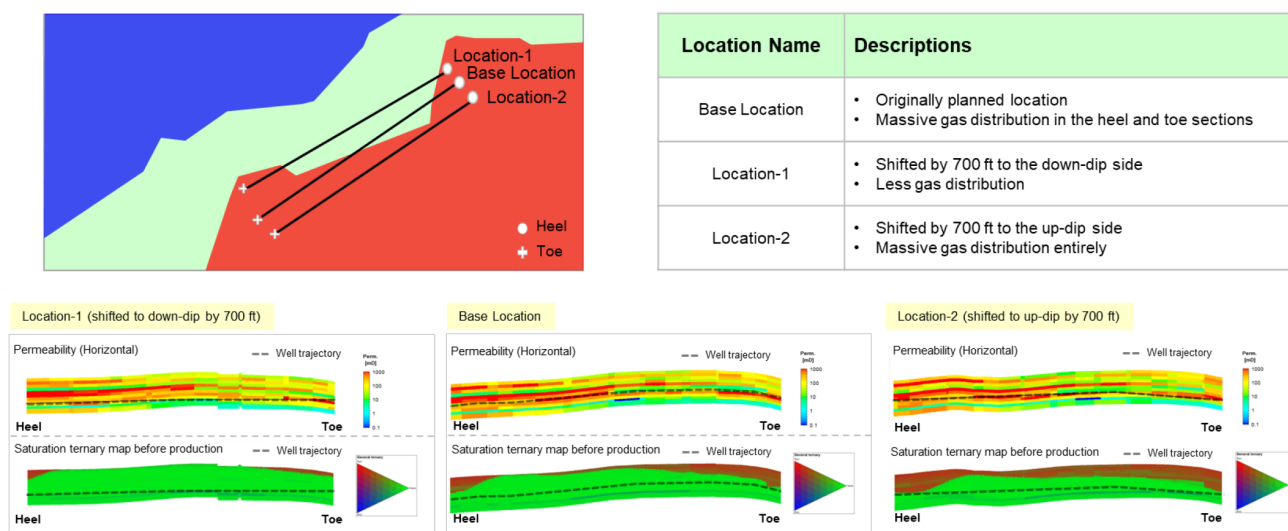


Figure 17—(Top) 3 well locations for sensitivity study and (Bottom) horizontal permeability and ternary saturation maps at well cross section.

Oil production and GOR trend for three locations with and without AICDs are summarized in Figure 18. At Location-1 with less gas distribution above Well-A before production starts, production gain by AICDs is the largest among three (3) cases with well life extension by 3 months. Location-2 shows more or less the similar oil production performance between No AICD and AICD cases. GOR increase trend is mitigated at Location-2 also, while the impact on shut-in timing is negligible because of the rapid increase in GOR.

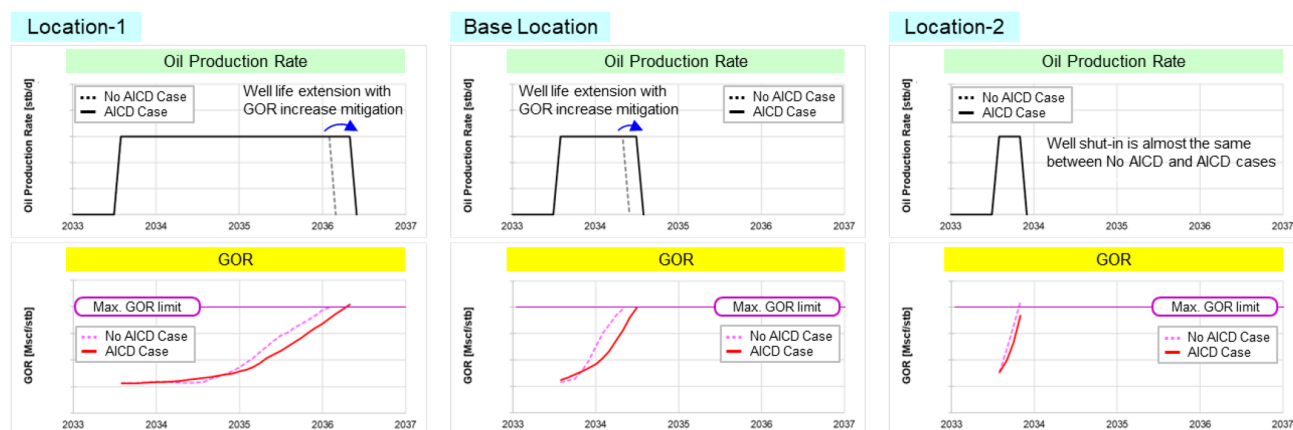


Figure 18—Well performance comparison among 3 location cases.

Figure 19 show sectional inflow performance and saturation maps at (1) Location-1 and (2) Location-2 respectively. There is no massive inflow section for No AICD case at Location-1, while inflow from low permeability section is improved by AICD application. At the end of production, higher GOR near the toe section was mitigated by AICD with more oil recovery from the latter half of the well. Oil saturation difference shows clear oil recovery enhancement in lower permeability zone. Even for Location-2, AICDs successfully built a uniform inflow distribution at the beginning as similar with Base Location and Location-1. GOR also increased uniformly in the entire well except for the heel section with high GOR from the beginning. Oil saturation difference also indicated the improvement in oil recovery from lower permeability zone, while the impact is much less than Location-1.

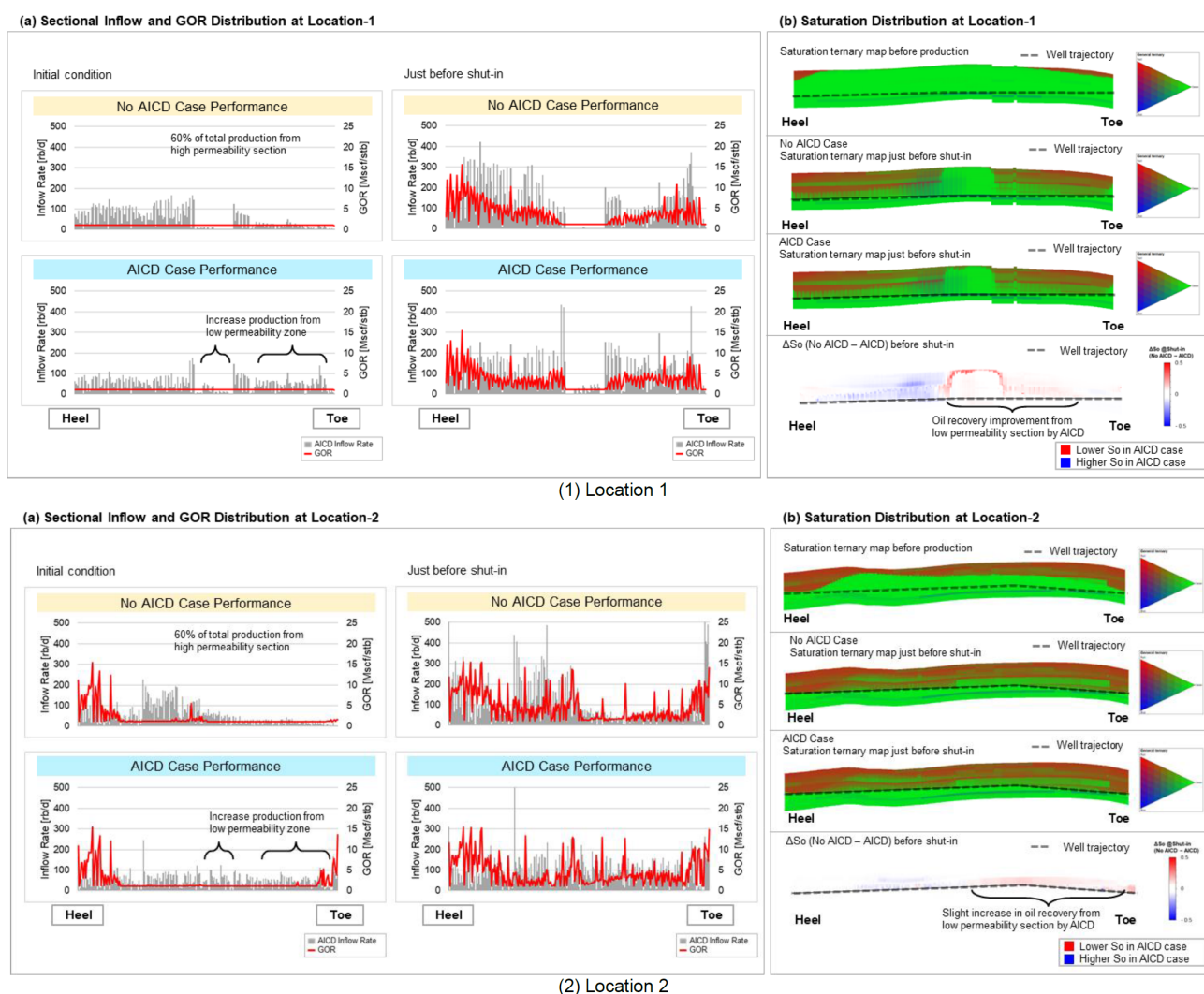


Figure 19—(a) Sectional inflow performance comparison between No AICD and AICD cases and (b) ternary saturation map for (1-upper) Location-1 and (2-lower) Location-2.

For all three (3) locations, oil recovery improvement from low permeability zone and uniform GOR increase in the entire well are observed in the model. However, the improvement in oil production performance is largely different among 3 locations. The gas breakthrough and GOR increase can be delayed under uniform gas flooding condition more than heterogeneous gas flooding. For Location-2 with massive gas distribution above Well-A, gas can shortly arrive Well-A followed by sharp increase in GOR regardless of uniform or heterogeneous flooding conditions. On the other hand, the gas breakthrough takes a longer time at Location-1 where there is less gas distribution above Well-A. The appropriate distance from gas front will enhance the effectiveness of AICDs and improve the well life.

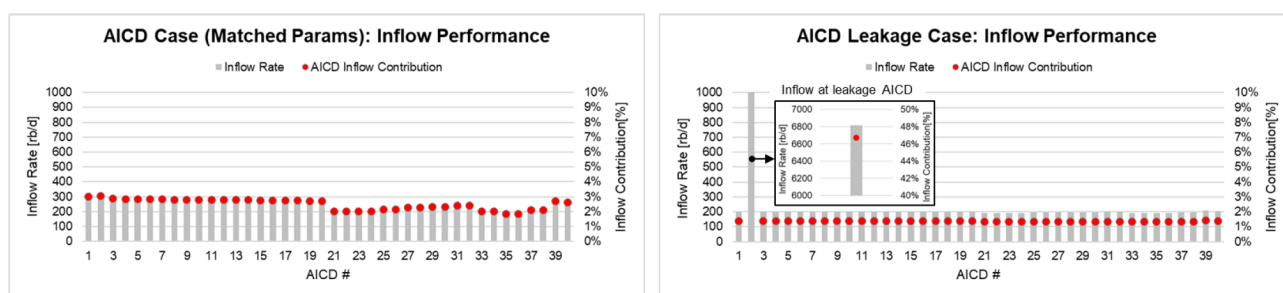
AICD leakage risk evaluation

A higher inflow rate compared with the designed nozzle performance curve was observed at all three (3) AICD wells during data review. During the history-matching of AICD well-2 performance, AICD leakage at higher inflow section also reproduced the well performance reasonably. The impact of AICD leakage on long-term production is a concern for AICD implementation and should be investigated. For leakage scenario, one (1) AICD out of 40 is assumed as a leaking one imposing extremely low pressure drop, while other AICDs are kept as designed parameter setting. The leaking AICD is deployed at the heel section in the simulation model as the worst scenario to accelerate gas coning. Table 5 shows the AICD nozzle performance parameters for leakage AICD scenario.

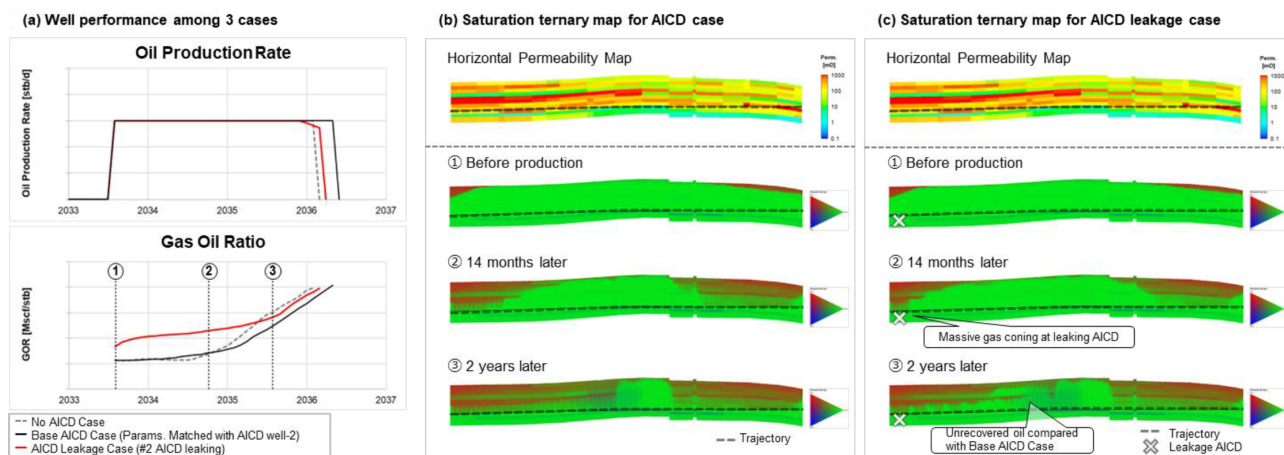
Table 5—AICD nozzle performance parameters for AICD leakage scenario.

Parameter (Unit)	Working AICD / Design values	Leaking AICD
a_{AICD} (bar/(kg/m ³)/(rm ³ /day) ^x)	1.3×10^{-5}	1.0×10^{-9}
x (-)	2.7	2.7
y (-)	1.0	1.0

For the evaluation of AICD leakage impact, Well-A at Location-1 is used. Figure 20(1) shows the comparison of AICD inflow performance at the initial condition between AICD case with matched parameters for AICD well-2 and AICD leakage case. AICD case with matched parameter shows uniform inflow profile from each AICD. For AICD leakage case, the leaking AICD allows excessive inflow by almost 50% of total production rate, while other AICDs show uniform performance. Long-term production performance showed a clear difference in GOR trend and resultant well life as shown in Figure 20(2). AICD leakage case shows higher GOR from the beginning due to the acceleration in gas breakthrough at the heel section.



(1) Inflow distribution for AICD case and AICD leakage case at the initial condition



(2) (a) Well performance comparison among No AICD, AICD and AICD leakage cases, (b) time-lapse cross-sectional saturation map for AICD case and (c) time-lapse cross-sectional saturation map for AICD leakage case.

Figure 20—Summary of well performance evaluation of the AICD leakage case.

AICD leakage is a reasonable risk and will have a negative impact on production performance. However, it can be mitigated with preventive and post-remedial actions. As corrosion is one of the concerns for AICD malfunction, corrosion resistance alloy application needs to be considered as a base mitigation action. Furthermore, erosion is also a common concern for the application of ICDs and AICDs, while the risk can be mitigated by deploying enough number of devices to manage inflow rate under erosional velocity. In case leakage is confirmed by PLT measurement, post-remedial actions need to be implemented. Casing patch

is an effective tool to isolate the leakage section, while it becomes choking beyond the isolated section. Squeezing cement is also a countermeasure, though there is a risk of damage or choking to other remaining sections.

Conclusion

This study covers the throughout contents on AICD evaluation in Reservoir-A consisting of: (i) exhaustive performance review of the existing AICD wells; (ii) the AICD nozzle performance calibration through history-matching; and (iii) the AICD effectiveness evaluation for the future planned well. The findings are summarized as below:

- The choking effect by AICDs was confirmed by the change of inflow performance during PLT survey after adding perforations at AICD locations. The inflow increased by maximum 6 times before additional perforation.
- The calibration of AICD nozzle performance model was successfully conducted by the integrated approach with steady-state wellbore simulation to reproduce inflow performance during PLT survey and dynamic reservoir simulation for long-term well performance matching. AICD parameter AICD parameter x adjustment case and leakage scenario reasonably reproduced the historical well performance, while AICD parameter y is still not estimated due to the limitation of data.
- The future well performance was evaluated under the gas flooding assumption with the calibrated AICD parameters. For single well model evaluation with hypothetical horizontal gas flooding assumption, well life extension was expected by AICDs application. AICDs will mitigate the excessive gas flooding through high permeability section.
- For AICDs application in the future well in the full field reservoir model, AICD shows a positive impact on well life. The inflow equalization and uniform gas flooding can be expected by AICDs, while selective gas regulation cannot be effective for the fluid properties in Reservoir-A. Furthermore, AICD effectiveness will be maximized when the well is located at the appropriate distance from the gas front.
- AICD leakage can be a reasonable risk based on the history matching for inflow profile during PLT survey. The leaking AICD will cause excessive inflow with gas coning acceleration and deteriorate the well performance. The applicability of preventive and post-remedial actions is to be investigated in advance.

Acknowledgement

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Nomenclature

A	= nozzle area, L^2 , ft^2 [m^2]
a_{ICD}	= ICD strength parameter, $psi/(RB/D)^2$ [$bar/(res\ m^3/d)^2$]
a_{AICD}	= AICD strength parameter, $psi/(lbm/ft^3)/(RB/D)^x$ [$bar/(kg/m^3)/(res\ m^3/d)^x$]
C_d	= Discharge coefficient, dimensionless
p_{res}	= reservoir pressure, $m/Lt^2\ psia$ [Pa]
q	= local volumetric flow rate, L^3/t , RB/D [$res\ m^3/d$]
x	= flow exponent, dimensionless
y	= viscosity exponent, dimensionless
μ	= fluid viscosity, m/Lt , cP [$Pa \cdot s$]

ρ = fluid density, m/Lt², lbm/ft³ [kg/m³]

Subscripts

cal = calibration fluid
mix = multiphase mixture
gas = gas property
oil = oil property
water = water property

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